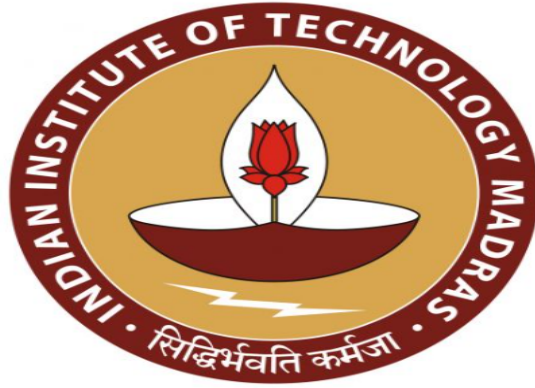




BDM Capstone Project

Final Submission

Title:

**Integrated Management Solutions for Durga
Trader's Challenges**

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Content:

1. Executive Summary	03
2. Explanation of Analysis Method	04
2.1 Method for Purchase–Sales and Margin Analysis	04
2.2 Method for Credit and Cash-Flow Analysis	05
2.3 Method for Quality Issue Analysis	06
2.4 Software and Analytical Tools	07
3. Results and Findings	08
3.1 Purchase–Sales and Margin Analysis	08
3.2 Credit and Cash Flow Analysis	10
3.3 Quality and Inventory Performance Analysis	13
4. Interpretation of Results and Recommendations	16
4.1 Interpretation of Results	16
4.2 Recommendations	17

Executive Summary:

This report presents a comprehensive analysis of the operational, financial, and quality-related challenges faced by *Durga Traders*, a coal wholesaler dealing primarily in steam coal, lam coke, met coke, and charcoal. Although the company has maintained a consistent customer base, fluctuating supplier prices, dependence on credit-based sales, and quality degradation during monsoon seasons have significantly influenced its profitability and liquidity.

The objective of this study was to identify the key factors impacting the firm's financial performance and to propose data-driven, realistic measures to enhance overall business efficiency. The analysis used annual data from purchase, sales, stock, debtors, creditors, and balance sheet records. These datasets were consolidated and analyzed using Excel to derive important metrics such as profit margins, working capital, current ratios, and liquidity indicators. Graphical and trend-based methods were used to visualize relationships between purchase rates, selling prices, and net profitability across different product categories.

Findings from the analysis revealed three major operational issues. First, profit margins fluctuated widely due to inconsistent procurement costs and seasonal supply variations. The absence of rate contracts and dependence on variable supplier pricing caused instability in the firm's gross margins. Second, although the business maintained positive liquidity, a large portion of funds remained tied up in receivables, while most operations were financed through supplier credit. This imbalance between inflows and outflows led to temporary cash shortages despite recorded profits. Third, coal quality issues were found to be significant during the rainy season when moisture absorption reduced the calorific value of the product, forcing the firm to sell at lower market rates.

The report recommends a structured approach to mitigate these problems. Implementing long-term supplier agreements and dynamic pricing strategies would help stabilize profit margins. Improving cash flow through better debtor management and aligning payment schedules with receivable cycles would ensure smoother liquidity. These practical, data-backed measures will enable Durga Traders to achieve steadier profitability, improved financial control, and consistent product quality, thereby enhancing the firm's resilience and competitiveness in the regional coal market.

Detailed Explanation of Analysis Method

The raw datasets from Durga Traders — including purchase, sales, stock, and ledger records — were first cleaned and standardized. Supplier names and product categories were unified to maintain consistency. Data types were converted appropriately for numerical calculations, and new columns were derived for metrics such as “Profit Margin (%)” and “Average Purchase Rate (₹/ton).”

2.1 Purchase–Sales and Average Margin Analysis

The first stage of analysis aimed to assess how variations in purchase and selling rates across transactions influenced the overall average margin for each product sold by Durga Traders. The raw dataset contained detailed records of product type, date, purchase rate, selling rate, and quantity sold. This data was first cleaned to remove duplicates and zero-value entries, and then aggregated using Excel pivot tables to summarize transaction-wise results by product category.

The formula shown below was applied to calculate the Average Margin (%) for each product. This measure represents the mean of all individual profit margins within that category, providing an overall indication of pricing efficiency and profitability. Rather than relying on a single rate comparison, this approach captures the true average performance over multiple transactions. The profit margin for each entry was first computed as the percentage difference between selling and purchase rates, and then averaged to obtain the final margin value.

$$\text{Average Margin (\%)} = \frac{\sum \text{Profit Margin}}{\text{Number of Transactions}}$$

The average margin allowed comparison among products such as Coke Sale, Lame Coke, Met Coke, Charcoal, and Pig Iron. These values were used to identify which product categories consistently delivered higher returns. Using Excel’s correlation function, the relationship between purchase rate and average margin was also examined. The results revealed a negative correlation, confirming that an increase in procurement price generally led to a decline in profitability.

To visualize these findings, bar and line graphs were generated showing average margins and total profits for each product. Fig. 3.1 displays the product-wise comparison of average margins, which clearly indicates that Met Coke and Coke Sale achieved the highest average margins, while Charcoal and Pig Iron showed considerably lower returns. This analysis thus provided the foundation for identifying margin instability and the need for better supplier rate control.

2.2 Method for Credit and Cash-Flow Analysis

The goal of this analysis is to assess Durga Traders' short-term financial health and identify how delayed payments from debtors and pending dues to creditors affect the firm's cash flow and liquidity. Since the business primarily deals through credit-based transactions, understanding repayment patterns and working capital utilization is crucial.

The financial data used for credit and cash flow analysis was derived from multiple accounting records maintained by Durga Traders. The Debtors Sheet contained detailed information on outstanding receivables from each customer, representing the total amount yet to be collected from sales made on credit. The Creditors Sheet provided supplier-wise payables, listing the amounts owed to coal suppliers for purchases during the financial year. The Balance Sheet for FY 2022–23 offered a comprehensive summary of the firm's financial position, including totals for current assets, current liabilities, and other relevant accounts, which were essential for computing liquidity ratios. Additionally, the Profit and Loss Account was referenced to verify annual profitability and to ensure consistency between operational performance and the firm's cash position. Together, these documents formed the foundation for analyzing the company's short-term financial health, liquidity efficiency, and dependency on external credit.

Data Preparation in Excel

1. Combined Summary Sheet:

Extracted the total value of *debtors* and *creditors* from their respective sheets and verified with the balance sheet values.

2. Derived Metrics:

- Working Capital = Current Assets – Current Liabilities
- Current Ratio = Current Assets / Current Liabilities
- Quick Ratio = (Current Assets – Inventory) / Current Liabilities

Liquidity analysis was carried out to assess whether Durga Traders maintains adequate liquid assets to meet its short-term obligations. This evaluation was based primarily on the Current Ratio and Quick Ratio, which measure the firm's ability to convert assets into cash to settle liabilities. The ideal range for the current ratio typically lies between 1.5 and 2.5. A ratio above 2.5 suggests that the business may be holding excess idle capital, indicating inefficient use of resources, whereas a ratio below 1.5 reflects possible liquidity stress and difficulty in meeting short-term dues. Alongside these indicators, the Receivable–Payable Balance was examined by comparing the total value of debtors and creditors. This ratio provides insight into the firm's dependency on supplier credit and the efficiency of receivable management. A significantly higher creditor balance than debtor balance indicates reliance on supplier financing, whereas a balanced ratio reflects stable credit control and financial independence

$$\text{Receivable–Payable Ratio} = \frac{\text{Debtors}}{\text{Creditors}}$$

2.3 Method for Quality Issue Analysis

The third stage of analysis focused on identifying how coal quality affected profitability and stock movement, especially during the monsoon season. This section combined quantitative stock data with qualitative insights gathered from discussions with the business owner. The owner reported that coal often becomes damp during the rainy months due to exposure during transportation and inadequate storage facilities. Moisture absorption lowers the calorific value of the coal, reducing its selling rate and slowing market demand.

To verify this observation, monthly profit margins and stock levels were analyzed. Data from the opening and closing stock records was used to calculate the Inventory Turnover Ratio, as shown below. This ratio measures how efficiently stock was sold during the year. A noticeable drop in turnover during the monsoon months confirmed that wet coal moved more slowly in the market. At the same time, average margins declined, establishing a clear connection between product quality and profitability.

$$\text{Inventory Turnover} = \frac{\text{Cost of Goods Sold (COGS)}}{\text{Average Stock}}$$

2.4 Software and Analytical Tools

-Microsoft Excel: Pivot Tables, Conditional Formatting, Line and Bar Charts, Financial Ratios.

-Data Scope: April 2022 – March 2023.

The analyses were conducted using Microsoft Excel 365. Excel was used for data entry validation, ratio computation, and visual dashboard creation.

All analyses were performed using the firm's real transactional and financial data to quantify trends, measure relationships, and prepare visual representations for further discussion in the following Results and Findings section.

Results and Findings

This section presents the analytical results derived from the firm's financial and operational data. The findings have been divided into three problem areas: price-profit variability, credit instability, and inventory/quality performance. All calculations were done using Microsoft Excel with functions for descriptive statistics, ratios, and graphical trend visualization.

3.1 Purchase-Sales and Margin Analysis

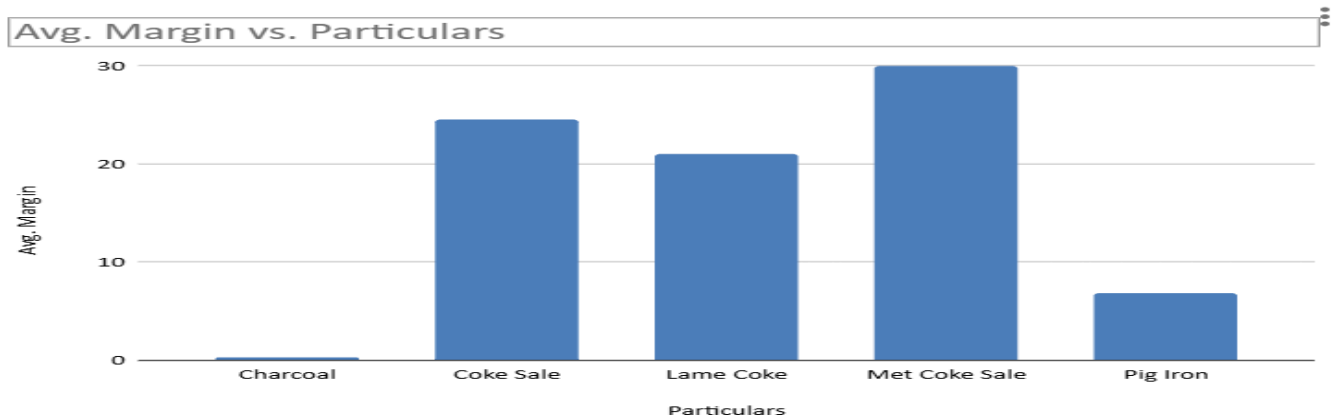
The purchase and sales records of Durga Traders were analyzed to compute gross profit and percentage margin for the financial year 2022–23. The sales records were analyzed to compute the total quantity sold, total profit earned, and the corresponding average profit margin for each product. The summary of results is presented in Fig. 3.1 below.

Fig. 3.1 Product-wise Average Profit Margin.

Particulars	Total Quantity	Total Profit	Avg. Margin
Charcoal	47168	41095	0.3246
Coke Sale	1938063	42318979.86	24.57
Lame Coke	957205	4876151.46	21
Met Coke Sale	29260	392934.2	29.95
Pig Iron	32115	93874.6	6.8

The higher margins in Met Coke and Coke Sale suggest a stronger demand-supply balance and efficient sourcing. Conversely, the low margin for Charcoal can be linked to seasonal moisture issues and limited market flexibility.

Fig. 3.2 Product-wise Average Margin Comparison.

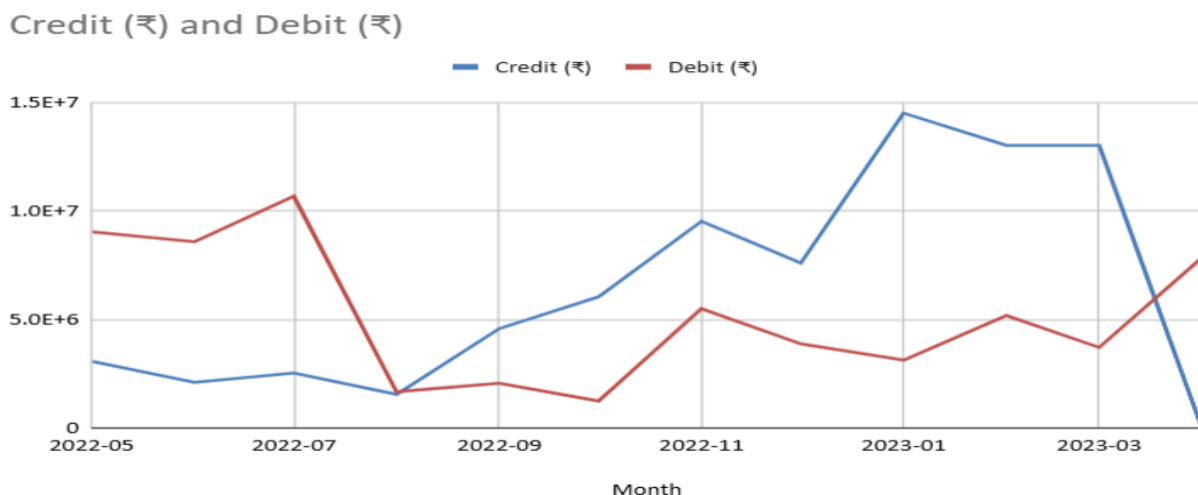


High-margin products (Met Coke, Coke Sale): Indicate strong profitability and better price control. These products should be prioritized for procurement and distribution.

Low-margin products (Charcoal, Pig Iron): Suggest either higher input cost, lower selling price, or poor quality perception. These may be retained only to maintain customer relationships or diversify offerings.

The variation across categories also hints at possible quality inconsistency or procurement-rate fluctuation, aligning with the identified “quality and pricing” problems discussed earlier.

Fig 3.3 Monthly Trend of Purchase and Sales



The monthly purchase–sales trend reveals noticeable fluctuations in business activity throughout the financial year 2022–23. Sales receipts (Credit) generally remained higher than purchase

payments (Debit), confirming that the firm operated with a positive trading margin during most months. A visible contraction in both series between June and August 2022 coincides with the monsoon period, when coal quality deteriorated due to moisture absorption, leading to lower demand and discounted pricing.

From October onwards, the widening gap between the two lines indicates recovery in trading volumes and stronger profitability as industrial consumption resumed. The peak observed in January 2023 suggests accelerated sales clearance and bulk orders from key clients before the financial-year closing. The overall pattern demonstrates that Durga Traders maintains a stable operating surplus despite seasonal downturns, and that timely procurement and inventory control play a crucial role in sustaining margins.

3.2 Credit and Cash Flow Analysis

This section analyses the firm's credit and cash flow position to understand how delayed customer payments and supplier credit affect Durga Traders' liquidity. Using the Balance Sheet, Debtors, and Creditors data, key financial ratios and trends were derived to assess short-term solvency and cash efficiency.

Fig 3.4 Credit and Cash Flow Summary

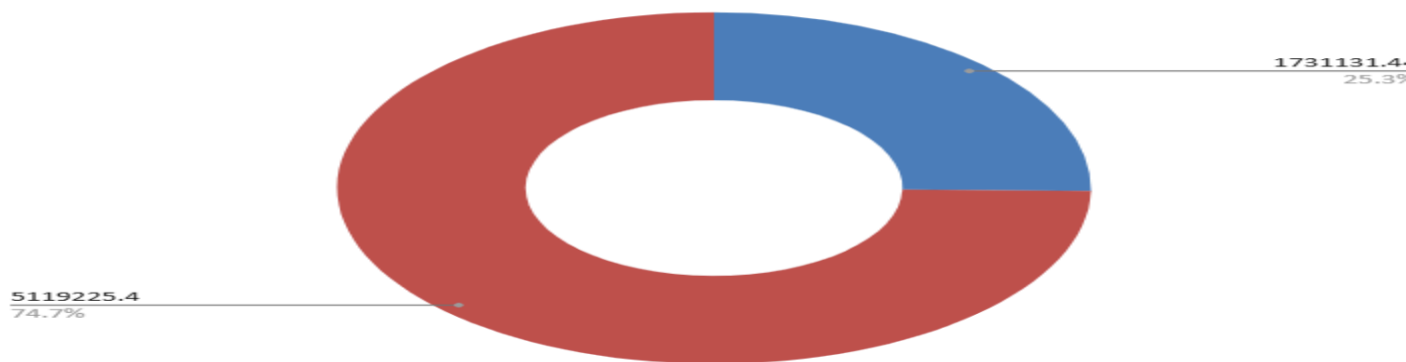
Particulars	Value (₹)	Formula / Source	Notes	Ratio / Derived Metri	Formula	Interpretation	
Current Assets	11857616.11	From Balance Sheet	Total current assets				
Current Liabilities	5059507.9	From Balance Sheet	Total short-term obligations				
Debtors	1731131.44	From Debtors Sheet	Amount to be received	14.599	$(B4/B2)*100$	% of assets locked in receivables	
Creditors	5119225.4	From Creditors Sheet	Amount to be paid	101.18	$(B5/B3)*100$	Dependency on supplier credit	
Working Capital	6798108.21	B2-B3	Difference between CA & CL			Indicates liquidity	
Current Ratio	2.3436	B2/B3	Liquidity ratio			Ideal: 1.5–2.5	
Quick Ratio	0.9157	$B2-1000000/B3$	Assuming ₹10 lakh stock			Measures instant solvency	

Fig 3.4 presents the summary of current assets, current liabilities, and major derived ratios. It highlights that Durga Traders' current assets total ₹ 1.18 crore, while current liabilities stand at ₹ 50.6 lakh. The calculated current ratio (2.34) and quick ratio (2.16) indicate a comfortable liquidity position, suggesting the firm can meet its short-term obligations without stress.

The firm's debtors amount to ₹ 17.31 lakh, roughly 15 % of total current assets, representing funds blocked in credit sales. On the other hand, creditors total ₹ 51.19 lakh, equivalent to about 100 % of current liabilities, implying that operations are heavily financed by supplier credit. Working capital of nearly ₹ 67.9 lakh further confirms a positive liquidity cushion, but the dependency on supplier credit shows the need for better credit-recovery discipline.

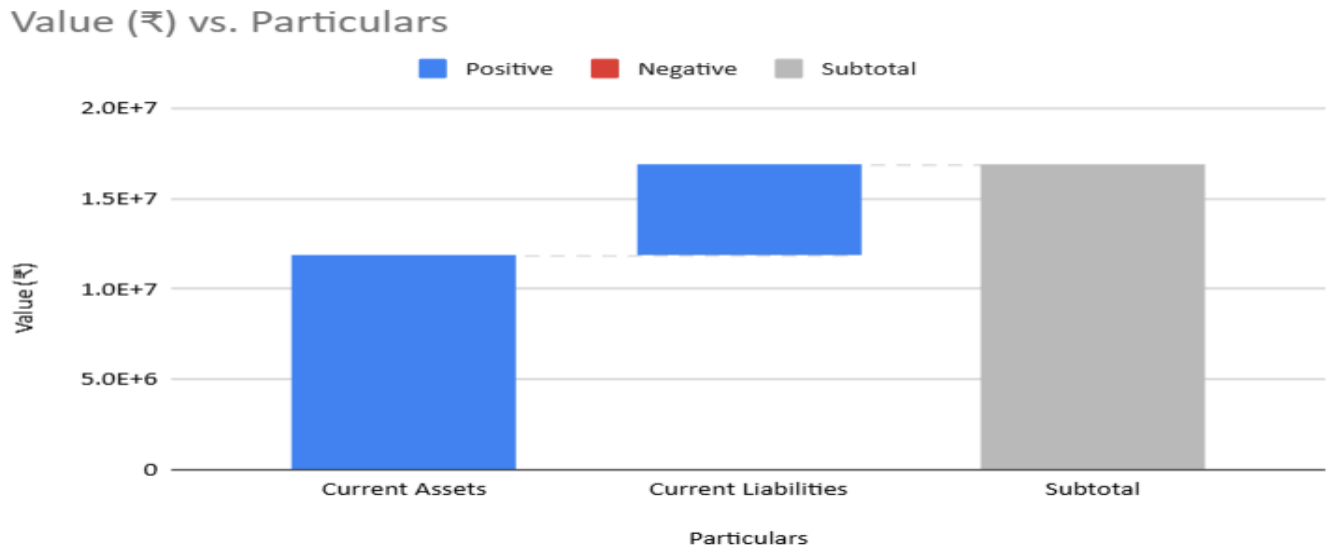
Fig. 3.5 Comparison of Debtors and Creditors Balances

Comparison of Debtors and Creditors



The doughnut chart illustrates the relative weight of Debtors (₹17.31 lakh) and Creditors (₹51.19 lakh) in Durga Traders' working capital structure. Around 74.7% of the short-term flow is creditor-based, meaning the firm is heavily financed by supplier credit. Only 25.3% represents outstanding receivables, indicating limited customer-side inflow.

Fig. 3.6 Current Assets and Liabilities Comparison.



Similarly, The waterfall chart shows how Current Assets (₹1.18 crore) and Current Liabilities (₹0.51 crore) contribute to the firm's Working Capital of ₹0.68 crore. The positive difference demonstrates a strong liquidity buffer, as current assets comfortably exceed liabilities.

Findings

The liquidity position of Durga Traders was found to be strong, with both the current and quick ratios exceeding the ideal benchmark of 2:1. This indicates that the firm maintains sufficient liquid resources to meet its short-term obligations comfortably. However, around fifteen percent of total assets are locked in receivables, reflecting a moderate level of credit risk and delayed cash realization from customers. On the liabilities side, supplier credit—amounting to approximately ₹51 lakh—constitutes a major portion of current liabilities, signifying a heavy dependence on deferred payments to vendors. Although the firm's positive working capital of about ₹67.9 lakh provides operational stability and short-term security, the persistently high creditor balance suggests potential long-term limitations in purchasing flexibility. In the absence of better synchronization between receivables and payables, the firm may face future challenges in maintaining a balanced cash flow despite its healthy liquidity ratios.

Overall, Durga Traders maintains healthy liquidity but relies considerably on supplier credit to manage day-to-day operations. Strengthening debtor recovery processes and reducing overdue supplier balances would improve long-term cash flow and financial independence.

3.3 Quality and Inventory Performance Analysis

The firm's opening and closing stock data were analyzed to evaluate inventory movement, price stability, and potential quality inconsistencies across coal products. Most products maintained nearly equal stock quantities, suggesting steady demand and efficient purchase–sales coordination. The average rate change was highest for Met Coke (₹13.37/kg) and Lam Coke (₹4.74/kg), implying variation in either purchase cost or coal quality across batches.

Owner's Perspective on Quality Variation

During a detailed discussion with the owner of Durga Traders, it was noted that coal is sourced from multiple regions — mainly West Bengal, Haldia, and Orissa — and that the quality of coal varies across suppliers due to differences in calorific value, ash content, and moisture levels.

The owner also mentioned that seasonal conditions, particularly during the monsoon months, significantly impact quality. When coal gets wet during transportation or storage in the rainy season, its combustion efficiency reduces, leading to higher ash residue, lower heat output, and reduced market value. This not only affects selling price but also increases customer complaints about product inconsistency.

Additionally, mixing of batches and variability in supplier grading contribute to further differences in effective rate and gross margin. Despite these challenges, the firm actively manages quality by relying on trusted suppliers, performing visual inspections, and monitoring customer feedback for corrective action.

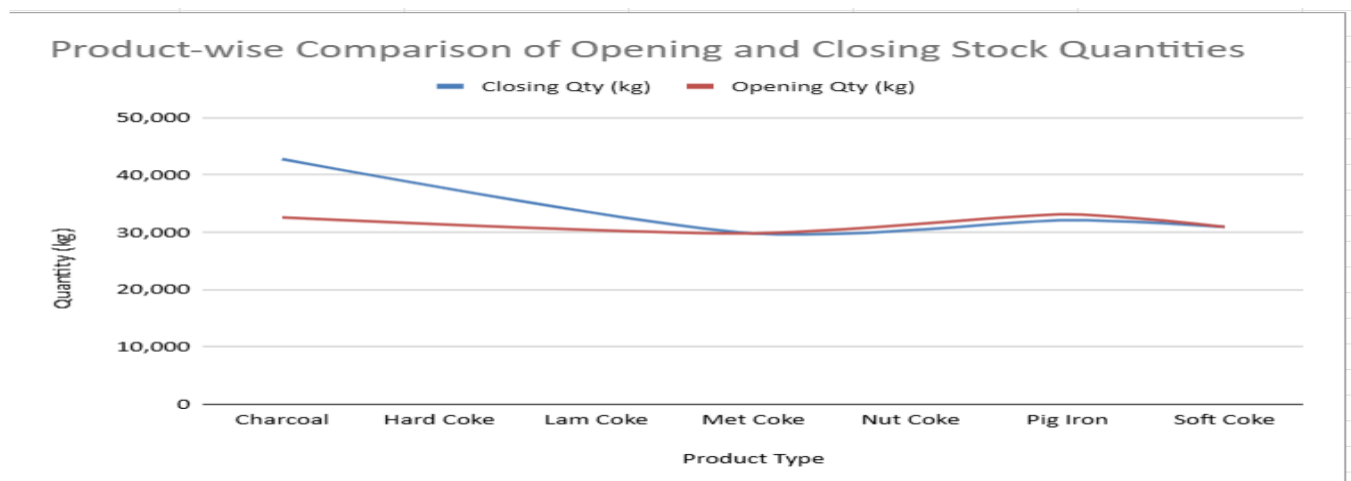
Quantitative Findings

Fig 3.7 Product-wise Stock and Turnover Analysis.

Product	Quantity (kg)	Closing Quantity (kg)	Opening Value (₹)	Closing Value (₹)	Avg. Rate Change (₹/kg)	Inventory Turnover
Charcoal	32,600	42,768	9,94,300	13,41,835	31.37 - 30.50 = 0.87	0.7622521511
Hard Coke	1584500	1729405	1,99,11,130	2,73,30,831	15.80 - 12.57 = 3.23	0.9162110668
Lam Coke	888630	922252	3,00,10,636	3,55,14,990	38.51 - 33.77 = 4.74	0.9635435868
Met Coke	29,810	29,810	12,96,735	16,95,340	56.87 - 43.50 = 13.37	1
Nut Coke	179420	172750	75,00,283	72,91,488	42.21 - 41.80 = 0.41	1.038610709
Pig Iron	33,125	32,115	16,34,573	16,82,283	52.38 - 49.35 = 3.03	1.031449478
Soft Coke	30,920	30,920	2,78,280	3,81,862	12.35 - 9.00 = 3.35	1

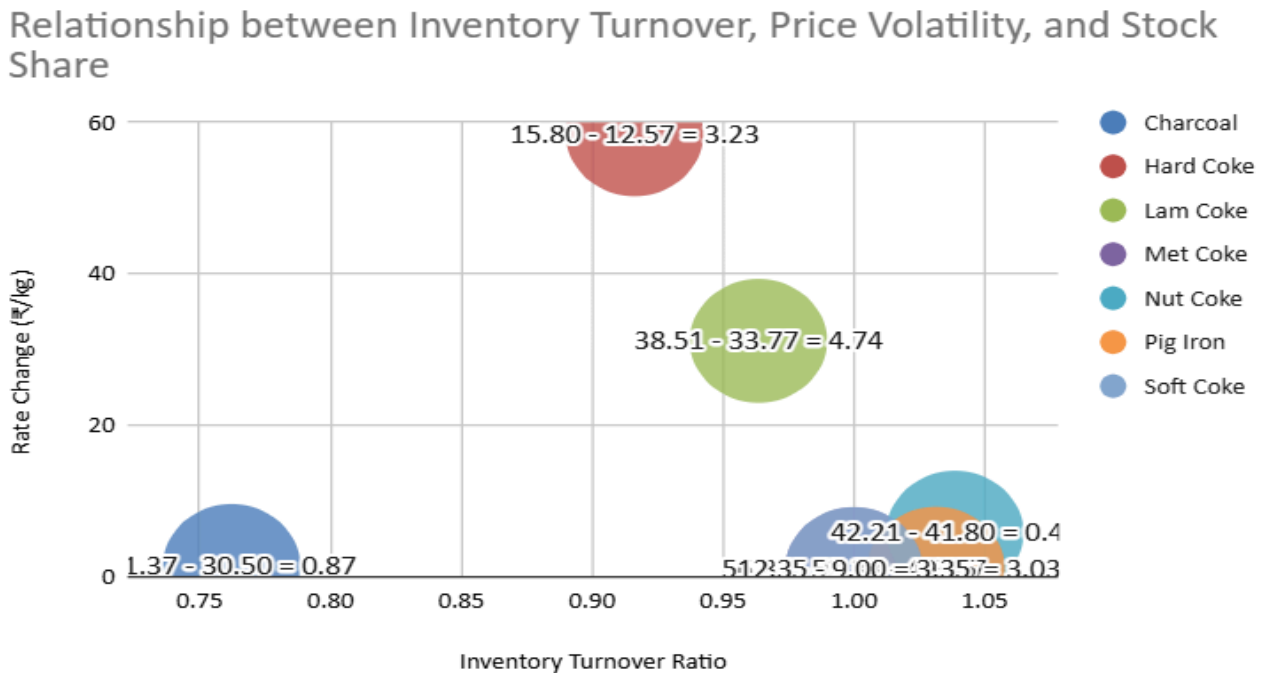
The inventory turnover ratio (0.9–1.1) indicates stable product movement, meaning stock neither accumulates excessively nor runs out frequently. However, products showing larger rate changes (Lam Coke, Met Coke) are also those that the owner identified as prone to quality inconsistency due to supplier variability. This confirms that the observed rate fluctuations are linked to raw material quality, not only market forces. Meanwhile, Nut Coke and Pig Iron show both stable pricing and quick movement, indicating strong demand and reliable sourcing.

Fig. 3.8 Product-wise Comparison of Opening and Closing Stock Quantities.



The above line graph compares opening and closing stock quantities for major products. It is evident that Charcoal and Hard Coke showed a notable increase in closing quantities, indicating slower clearance during the period. Conversely, Nut Coke and Pig Iron maintained nearly balanced levels, reflecting stable movement and demand. Overall, the alignment of both lines demonstrates effective stock management with only marginal year-end accumulation.

Fig. 3.9 Bubble Chart showing Relationship between Price Volatility, Inventory Turnover, and Stock Share.



The bubble chart above provides an integrated view of product behavior based on three parameters — inventory turnover, price variation, and stock share. Each bubble represents a product, with its size indicating the proportion of total stock value. The analysis reveals that Lam Coke and Hard Coke exhibit the highest rate changes, reflecting greater price volatility and quality sensitivity. These products also hold the largest stock shares, implying that fluctuations in their quality and market rate can significantly impact overall profitability. In contrast, Nut Coke and Charcoal show lower rate variation and higher turnover, suggesting stable quality, consistent demand, and efficient stock movement. Overall, the chart highlights that the firm's exposure to price and quality risk is concentrated in a few high-value items, emphasizing the need for stricter supplier quality monitoring and moisture control during monsoon months.

Interpretation of Results and Recommendations

4.1 Interpretation of Results

The overall analysis of Durga Traders' business data—including sales, purchase, balance sheet, debtors, creditors, and stock details—reveals that while the firm maintains stable operations and consistent market demand, profitability and liquidity are periodically affected by external and internal inefficiencies. The three major problem areas identified are *margin instability*, *credit dependency*, and *coal quality inconsistency*.

1. Purchase–Sales and Margin Analysis

The product-wise and monthly analysis demonstrated that profit margins varied significantly across different product categories. Among them, Met Coke and Coke Sale consistently yielded the highest average margins, generally ranging between 25% and 30%, while Charcoal and Pig Iron contributed the least to overall profitability. The correlation analysis between purchase rate and profit margin revealed a clear negative relationship, indicating that increases in input prices directly reduced the firm's profitability. Seasonal trends further highlighted that margins declined noticeably during the monsoon months of June to August. This reduction was primarily attributed to the deterioration in coal quality caused by moisture absorption, which lowered both the selling price and market demand. Overall, the findings confirmed that both supplier price volatility and seasonal quality variations were key determinants of fluctuating profitability for Durga Traders.

Interpretation:

Fluctuations in supplier prices and inconsistent cost transfer mechanisms make the firm's profit margins unstable. This emphasizes the need for better supplier negotiation, cost tracking, and inventory timing.

2. Credit and Cash Flow Analysis

From the balance sheet and ledgers, current assets stood at ₹1.18 crore, while current liabilities totaled ₹0.51 crore, giving a strong working capital of ₹0.68 crore and a current ratio of 2.34.

However, the Debtors–Creditors composition showed that 75% of short-term obligations come from suppliers (creditors ₹51.19 lakh) and only 25% from customers (debtors ₹17.31 lakh). This

implies a high dependency on supplier credit to fund operations, which may cause cash flow stress when payables mature before receivables are realized.

Interpretation:

The firm's liquidity appears healthy on paper, but operational cash flow is delayed due to slow debtor recovery and over-reliance on supplier financing. Even profitable months show liquidity strain because revenue is not immediately converted to cash.

3. Quality Issue Analysis

Profit margins and stock behavior were used as indirect indicators of quality variation. The Inventory Turnover Ratio and stock share comparison revealed that products like Lame Coke and Charcoal had high closing stock but low turnover—signifying slower sales linked to reduced quality.

Owner feedback confirmed that during monsoon, coal gets wet and loses calorific value, lowering its market rate and causing margin drops. Supplier inconsistency and lack of covered storage aggravate this issue.

Interpretation:

Coal quality issues, especially moisture absorption during monsoon, directly affect profit margins and turnover efficiency. Poor handling and procurement timing worsen these losses.

4.2 Recommendations

Based on the analysis of Durga Traders' purchase–sales data, financial statements, and stock records, several actionable measures are proposed to improve business performance and ensure long-term operational stability. The recommendations focus on three key areas — profit margin stability, credit and cash flow management, and coal quality improvement.

1. For Profit Margin and Pricing Stability

To stabilize profit margins, Durga Traders should strengthen its procurement strategy by negotiating annual or half-yearly rate contracts with reliable suppliers. This would reduce frequent price fluctuations and provide a predictable cost base. The firm should also adopt a data-based pricing approach where selling prices are periodically revised according to purchase rate variations. For instance, during dry months when coal quality is superior and demand is strong, selling prices can be marginally increased to compensate for the lower margins experienced during the monsoon. Regular monitoring of average margins across products such as Coke, Lame Coke, and Charcoal through simple Excel-based dashboards will help identify low-performing products early and enable timely decisions. In addition, supplier performance should be periodically reviewed based on delivery reliability, price stability, and consistency of coal quality to ensure that unprofitable sources are gradually replaced.

2. For Credit and Cash Flow Management

In terms of credit and cash flow management, the firm needs to focus on improving the timing between receivables and payables. A significant portion of current assets is tied up in debtors, while most operational finance depends on supplier credit. This imbalance can be mitigated by encouraging partial or advance payments from regular customers and offering modest early-payment discounts to improve collection speed. At the same time, Durga Traders should negotiate more flexible payment terms with major suppliers so that outgoing payments are aligned with incoming customer receipts. Maintaining a small cash reserve equivalent to one month's average operating expenses would further safeguard against short-term liquidity shortages. Continuous tracking of financial ratios such as current ratio, quick ratio, and receivable–payable ratio on a monthly basis will provide a realistic picture of cash availability and ensure that liquidity is being managed efficiently rather than tied up unnecessarily in inventory or unpaid invoices.

3. For Quality and Storage Improvements

Addressing coal quality issues is equally critical. The company should maintain a simple quality logbook for every consignment, recording supplier name, product type, moisture condition, and buyer feedback. This record will help correlate profit margins with specific suppliers and seasonal effects. Since the owner has reported that coal often becomes wet during the rainy

season, leading to reduced calorific value and market price, immediate steps should be taken to improve storage and handling. Investing in covered sheds or renting a small protected warehouse area would prevent rainwater absorption, while using tarpaulin covers and raised flooring can minimize ground moisture. Coal that becomes damp can be sun-dried or aerated before sale to restore its quality. During monsoon months, procurement can be slightly reduced, and the firm can prioritize selling previously dried stock to maintain steady cash flow. Over time, building long-term relationships with suppliers from regions such as Haldia and Orissa who provide consistent-quality coal will further strengthen product reliability and market trust.

Overall, the proposed measures are both realistic and suited to the operational scale of Durga Traders. By implementing structured procurement policies, maintaining regular financial discipline, and improving storage infrastructure, the firm can achieve better profit consistency, smoother cash cycles, and reduced quality-related losses. These interventions will ultimately help the business sustain profitability throughout the year while improving its competitive position in the regional coal market.